

APPLICATION DEVELOPMENT

CLINICAL TRIAL MANAGEMENT SYSTEM

Database Design Documentation

1. Database Design and Schema Documentation

This section documents the relational database design underlying the Clinical Trial Management System (CTMS) described in the accompanying Software Requirements Specification. It presents the entity-relationship model, the structure and sample contents of the core tables, the relationships and referential integrity rules that bind them, and a walkthrough of the normalization applied to the schema.

1.1 Entity Relationship (ER) Diagram

The diagram below shows the ten core entities of the CTMS schema and the associations between them. Studies sits at the center of the model: every participant, protocol visit, adverse event, investigational-product (IP) batch, protocol deviation, regulatory document, and audit entry is scoped to a specific study. Participants and Users, in turn, drive the transactional tables — eCRF Data, Adverse Events, and Protocol Deviations — that capture day-to-day trial activity.

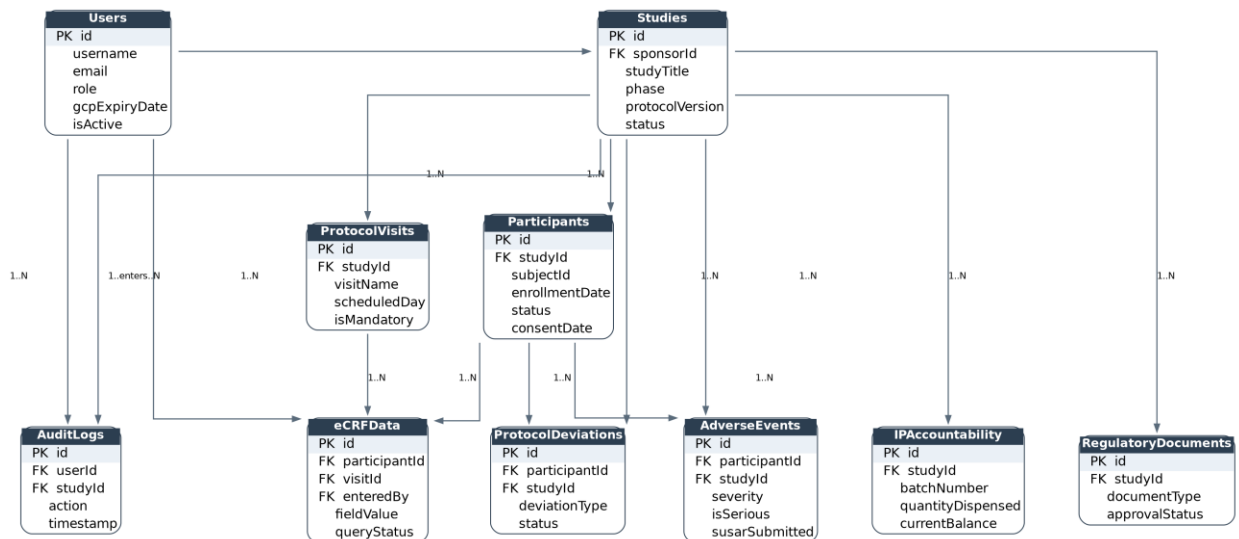


Figure 4.1 — CTMS Entity Relationship Diagram (crow's-foot cardinality shown on connectors)

Reading the Diagram

- PK denotes the primary key of the entity; FK denotes a foreign key referencing another entity's primary key.
- All 1..N associations originate from the “one” side (e.g., Studies) toward the “many” side (e.g., Participants).
- Users participates in two distinct roles: as the enterer of eCRF data and as the actor recorded in the Audit Logs.

2 Table Descriptions

The five subsections below describe the primary operational tables of the schema. Each table lists its columns with data types and constraints, followed by representative sample rows illustrating typical contents.

2.1 Users

Stores every research-staff account — administrators, sponsors, investigators, coordinators, data managers, and regulatory affairs officers — along with GCP training status and electronic-signature credentials required for 21 CFR Part 11 compliance.

Column	Type	Constraint
id	BIGINT	PK, AUTO_INCREMENT
username	VARCHAR(100)	UNIQUE, NOT NULL
email	VARCHAR(150)	UNIQUE, NOT NULL
passwordHash	VARCHAR(255)	NOT NULL
role	ENUM	NOT NULL
institutionalAffiliation	VARCHAR(200)	NOT NULL
gcpCertNumber	VARCHAR(50)	NOT NULL
gcpExpiryDate	DATE	NOT NULL
electronicSignatureHash	VARCHAR(255)	NOT NULL
isActive	BOOLEAN	DEFAULT TRUE

Sample Data

Id	Username	Role	Institutional affiliation	Gcp expiry date
101	dr.a.rao	PRINCIPAL_INVESTIGATOR	Apollo Research Institute	2027-03-15
102	s.mehta	SITE_COORDINATOR	Apollo Research Institute	2026-11-02
103	n.iyer	DATA_MANAGER	CTMS Sponsor Corp.	2027-01-20

2.2 Studies

Represents a single clinical trial protocol, including its phase, sponsor, therapeutic area, protocol version, and enrollment targets.

Column	Type	Constraint
id	BIGINT	PK, AUTO_INCREMENT
studyTitle	VARCHAR(500)	NOT NULL
sponsorId	BIGINT	FK → Users.id
phase	ENUM	NOT NULL
therapeuticArea	VARCHAR(100)	NOT NULL
protocolVersion	VARCHAR(20)	NOT NULL
status	ENUM	NOT NULL
targetEnrollment	INT	NOT NULL
irbApprovalDate	DATE	NULLABLE

Sample Data

Id	Study title	Phase	Protocol version	Status
501	Efficacy of Compound X in T2 Diabetes	PHASE_2	v2.1	ACTIVE
502	Long-term Safety of Biologic Y	PHASE_3	v1.4	ACTIVE
503	Observational Study — Cardiac Outcomes	OBSERVATIONAL	v1.0	SETUP

2.3 Participants

Stores pseudonymized subject records linking an enrolled participant to a study and site, preserving blinding through encrypted arm assignment.

Column	Type	Constraint
id	BIGINT	PK, AUTO_INCREMENT
studyId	BIGINT	FK → Studies.id
subjectId	VARCHAR(50)	UNIQUE, NOT NULL
siteId	BIGINT	FK → Sites.id
enrollmentDate	DATE	NOT NULL
armAssignment	VARCHAR(50)	Encrypted, blinded

Column	Type	Constraint
status	ENUM	NOT NULL
consentVersion	VARCHAR(20)	NOT NULL
consentDate	DATE	NOT NULL

Sample Data

Id	Study ID	Subject ID	Status	Consent date
9001	501	CTX-0142	ENROLLED	2026-02-10
9002	501	CTX-0143	ACTIVE	2026-02-11
9003	502	BLY-0057	SCREEN_FAILED	2026-01-30

2.4 eCRF Data

Captures every electronic case report form field entry against a participant's scheduled visit, with a full audit history of modifications per 21 CFR Part 11.

Column	Type	Constraint
id	BIGINT	PK, AUTO_INCREMENT
participantId	BIGINT	FK → Participants.id
visitId	BIGINT	FK → ProtocolVisits.id
fieldId	VARCHAR(100)	NOT NULL
fieldValue	TEXT	—
enteredBy	BIGINT	FK → Users.id
enteredAt	TIMESTAMP	NOT NULL
modificationReason	TEXT	Required if modified
queryStatus	ENUM	DEFAULT 'NONE'

Sample Data

Id	Participant	Field ID	Field value	Query status
70021	9001	vitals_systolic_bp	128	NONE
70022	9001	vitals_diastolic_bp	82	OPEN
70023	9002	lab_glucose_fasting	104	CLOSED

2.5 Adverse Events

Records every adverse event reported against a participant, including severity, causality, and whether the event meets seriousness criteria requiring expedited SUSAR submission.

Column	Type	Constraint
id	BIGINT	PK, AUTO_INCREMENT
participantId	BIGINT	FK → Participants.id
studyId	BIGINT	FK → Studies.id
onsetDate	DATE	NOT NULL
severity	ENUM	NOT NULL
causality	ENUM	NOT NULL
isSerious	BOOLEAN	NOT NULL
reportedBy	BIGINT	FK → Users.id
susarSubmitted	BOOLEAN	DEFAULT FALSE

Sample Data

id	participantId	severity	isSerious	susarSubmitted
30011	9001	MODERATE	FALSE	FALSE
30012	9002	SEVERE	TRUE	TRUE
30013	9003	MILD	FALSE	FALSE

3. Database Relationships

The schema is organized around a strict hierarchy: a Study owns its Participants, Protocol Visits, Adverse Events, IP Accountability records, Protocol Deviations, Regulatory Documents, and Audit Logs. Users act as the actor across several of these relationships — as sponsor of a Study, as the person entering eCRF data, and as the actor recorded in every audit event.

Parent Table	Child Table	Cardinality	Description
Users	Studies	1 : N	A sponsor user oversees one or more studies
Studies	Participants	1 : N	A study enrolls many participants
Studies	Protocol Visits	1 : N	A study defines its own visit schedule
Participants	eCRF Data	1 : N	Each participant has many eCRF field entries
Protocol Visits	eCRF Data	1 : N	Each visit is linked to many form entries
Participants	Adverse Events	1 : N	A participant may report multiple AEs
Studies	IP Accountability	1 : N	A study tracks many IP batches
Participants	Protocol Deviations	1 : N	A participant may have multiple deviations
Studies	Regulatory Documents	1 : N	A study accumulates many TMF documents
Users	Audit Logs	1 : N	Every user action generates an audit record

No many-to-many relationships exist at the core-entity level; where a conceptual many-to-many arises — for example, a Site Coordinator supporting several Studies — it is resolved through a join table (Study Staff Assignment) carrying the delegation record required for GCP-compliant access control.

4. Referential Integrity Rules

- Every foreign key (studyId, participantId, visitId, sponsorId, enteredBy, reportedBy, userId) must reference an existing row in its parent table; orphaned child rows are rejected at the application and database layer.
- Deletion of a Study is restricted (ON DELETE RESTRICT) while any Participants, eCRFData, or AdverseEvents rows reference it, preventing accidental loss of regulated trial data.
- Participant deactivation (status = WITHDRAWN) is a soft delete; the row and its full history remain intact to satisfy the 15-year data-retention rule.
- eCRFData rows are never physically deleted after entry — corrections are recorded as new audit-trailed values with modificationReason populated, preserving the original value per 21 CFR Part 11.
- AuditLogs rows are append-only: no UPDATE or DELETE operation is permitted against this table by any role, including System Administrator.
- A Participant cannot be created without a valid studyId, and enrollmentDate must fall on or after the referenced Study's irbApprovalDate.
- An AdverseEvent's participantId and studyId must belong to the same Study record — cross-study references are rejected by a composite-key check constraint.
- Unique constraints on Users.username, Users.email, and Participants.subjectId prevent duplicate identity records across the platform.

4.1 Database Normalization (1NF, 2NF, 3NF)

The schema was normalized to third normal form (3NF) to eliminate redundancy, protect data integrity, and keep the audit trail unambiguous — all essential properties for a GCP-regulated system.

First Normal Form (1NF)

Every table stores atomic values only: a Participant's identifying data and clinical arm assignment are separate columns rather than a combined free-text field, and eCRFData stores exactly one field value per row rather than embedding multiple form answers in a single cell. Repeating groups — such as a participant's multiple adverse events — are broken out into their own Adverse Events table rather than repeated columns (ae1, ae2, ae3) inside Participants.

Second Normal Form (2NF)

All tables use a single-column surrogate primary key (id), so partial-key dependency cannot occur; every non-key column depends on the whole key. For example, in eCRFData, fieldValue and enteredAt depend on the full combination of participant, visit, and field context, which is why the design captures that context through foreign keys rather than compressing it into a composite key that a partial dependency could violate.

Third Normal Form (3NF)

Transitive dependencies were removed by extracting reference data into its own tables. institutionalAffiliation-level detail lives in Users rather than being copied onto every Study row; a Study's sponsor details are reached through sponsorId rather than duplicated in the Studies table. Similarly, enumerated classification values (severity, causality, phase, status) are constrained through ENUM types rather than descriptive text repeated across rows, and site-specific facts are isolated in a Sites table referenced by siteId rather than embedded in Participants.

Why 3NF (and Not Higher)

- 3NF removes practically all update, insertion, and deletion anomalies relevant to this schema while keeping query joins manageable for real-time protocol-deviation detection (< 1 second target).
- Selective denormalization is intentionally applied to Studies.actualEnrollment, a maintained counter cache, to keep the enrollment dashboard responsive without recomputing counts across millions of eCRFData rows on every page load.
- Higher normal forms (BCNF, 4NF) were evaluated but offered no additional integrity benefit here, since no table currently exhibits multi-valued or join dependencies beyond 3NF.

5 Database Design Summary

The CTMS schema is a Study-centric relational model comprising ten core tables — Users, Studies, Participants, Protocol Visits, eCRF Data, Adverse Events, IP Accountability, Protocol Deviations, Regulatory Documents, and Audit Logs — connected through well-defined one-to-many relationships and enforced foreign-key constraints.

- The design is normalized to 3NF, minimizing redundancy while keeping the audit trail complete and tamper-evident.
- Referential integrity rules protect against orphaned records and enforce GCP-mandated retention and immutability requirements.
- A single, targeted denormalization (enrollment counter caching) balances integrity against the system's sub-second performance targets.
- The model directly supports the functional requirements defined in the SRS: real-time protocol-deviation detection, 24-hour SAE escalation, 21 CFR Part 11 electronic signatures, and a searchable, GCP-compliant audit trail.

Taken together, the ER diagram, table definitions, relationship map, and normalization rationale in this section give implementers a complete and unambiguous blueprint for building the CTMS persistence layer.